

Demonstration of GazeNoter: Enhancing AR Note-Taking Through Gaze-Based Selection of LLM Suggestions

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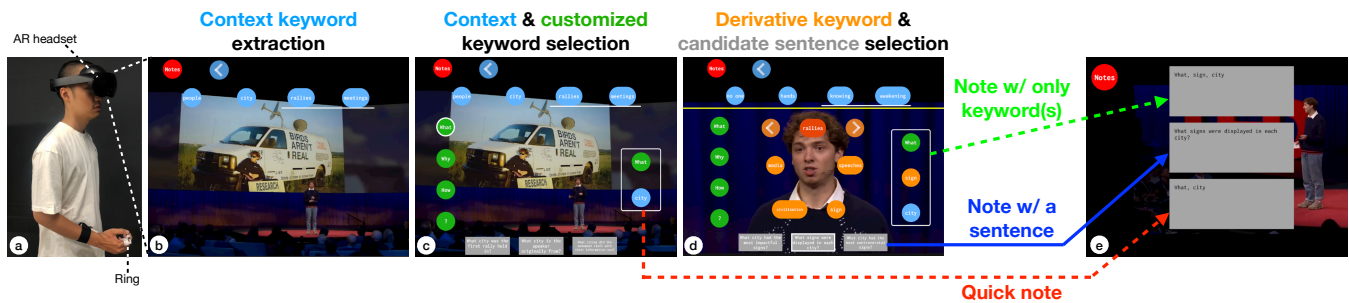


Figure 1: Users swiftly select LLM-generated suggestions via gaze on an AR headset with a ring for real-time note-taking in GazeNoter. (a) An AR headset and a ring are worn for note-taking in a speech. (b) Extracted context keywords from the latest sentence of the speech. (c) Users select context and customized keywords, and candidate sentences are automatically generated. (d) Users select derivative keywords beyond the context of the speech and select a candidate sentence best matching the intentions to record as a note. (e) Users review the recorded notes. (Blue) Normally, a sentence is recorded as a note. (Green) If no candidate sentences match users' intentions, users could record all selected keywords as a note. (Red) If users need to take a note hastily, users could select context (or also customized) keywords, skipping step (d), to record these as a *quick note*.

Abstract

Note-taking is critical during speeches and discussions, serving not only for later summarization and organization but also for real-time question and opinion reminding in question-and-answer sessions or timely contributions in discussions. Manually typing on smartphones for note-taking could be distracting and increase cognitive load for users. While large language models (LLMs) are used to automatically generate summaries and highlights, the content generated by artificial intelligence (AI) may not match users' intentions without user input or interaction. Therefore, we propose

an AI-copiloted augmented reality (AR) system, GazeNoter, to allow users to swiftly select diverse LLM-generated suggestions via gaze on an AR headset for real-time note-taking. GazeNoter leverages an AR headset as a medium for users to swiftly adjust the LLM output to match their intentions, forming a user-in-the-loop AI system for both within-context and beyond-context notes. We conducted two user studies to verify the usability of GazeNoter in attending speeches in a static sitting condition and walking meetings and discussions in a mobile walking condition, respectively.

CCS Concepts

• **Human-centered computing** → **Mixed / augmented reality**; *Natural language interfaces*; *Text input*.

Keywords

note-taking, AR, LLM, AI, gaze input, wearable devices

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1 Introduction

Note-taking is crucial during speech-based activities, such as in speeches and discussions. However, manual note-taking can divert attention from the primary activity, distracting users and increasing their cognitive load [5, 6]. Prior research [4, 7, 8, 11] has utilized natural language processing (NLP) and large language models (LLMs) to address these challenges. However, the auto-generated notes may not always align with the users' intentions. To tackle this issue, recent studies [2, 3] have sought to involve user input in the automatic note-taking process by enabling users to highlight crucial statements during meetings. Nevertheless, these methods still fall short as they generate notes only within the context of the transcript but cannot produce notes or inferences that go beyond the context.

A nuanced system that incorporates user input into automatic note-taking, ensuring the alignment of generated content with users' intentions, and providing support for both within-context and beyond-context notes is then required. By integrating user input into AI, forming a user-in-the-loop interaction paradigm, users can copilot with AI to adjust and generate more precise and desirable outcomes. Implementing this user-in-the-loop AI system for real-time note-taking requires a medium that facilitates swift, subtle, and low-distraction input and output manners for displaying and adjusting LLM-generated content to reduce users' cognitive load and match users' intentions. Augmented reality (AR) headsets with see-through displays, offering gaze selection, are ideal for this purpose, reducing the need to look down at smartphones and minimizing distraction.

We present an AI-copilot AR system, GazeNoter, enabling users to swiftly select LLM-generated suggestions via gaze on an AR headset for real-time note-taking. GazeNoter extracts keywords from the context of real-time audio transcripts via the LLM. Users can select these context keywords, or prompt the LLM to derive more keywords from the selected context keywords for further selection, aligning closely with users' intentions. Using the selected keywords, the LLM then organizes candidate sentences that might encapsulate the desired notes for users to record the notes. By leveraging the three capabilities of the LLM, including extraction, derivation and organization, and efficient gaze selection for keyword and sentence selection, this AI-copilot AR system achieves real-time note-taking with low distraction and cognitive load, matching users' intentions for both within-context and beyond-context notes. While individual components, including LLMs for note-taking and AR gaze interactions, have been explored before, GazeNoter uniquely integrates these for mobile, real-time note-taking without voice or typing input. We demonstrate GazeNoter in two scenarios: (1) attending speeches in a static sitting condition; and (2) walking meetings and discussions in a mobile walking condition.

2 GazeNoter

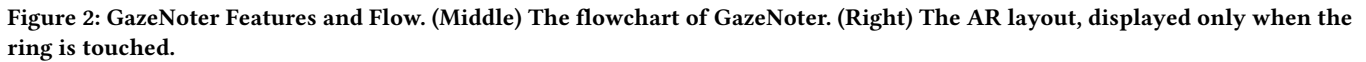
We propose GazeNoter [9] for real-time note-taking with low distraction and cognitive load, and matching users' intentions in speech-based activities for within-context and beyond-context notes.

GazeNoter showcases the LLM-generated suggestions on the AR headset, enabling users to select and adjust these contents via gaze and ring input to take their desired notes. Users select LLM-generated keywords and then select an LLM-generated sentence to record as a note, as shown in Figure 2. In this keyword and sentence selection procedure, four essential features are leveraged, including the selection of three types of keywords, *context keyword*, *customized keyword*, and *derivative keyword*, along with the selection of *candidate sentence*. Furthermore, one additional feature, *note refinement*, as well as these four features constitute GazeNoter. All AR contents are displayed on the AR headset only when the button of the ring is touched for note-taking. Users utilize gaze to select an item and click the button to confirm the selected item. Double-click is used as an alternative selection method. We designed the ring based on the comfort zone in [10].

For the context keyword selection, the system captures a voice segment, subsequently displaying context keywords via the LLM keyword extraction in real-time from the latest sentence. As shown in Figure 2 (a), these context keywords appear on the top section of the AR display in a 1×4 layout, at most 4 keywords from the most recent sentence added in the last of the queue clearly underlined. Users can also view previous context keywords using the previous/next arrow buttons. Users select one or multiple context keyword(s) relevant to the content they want to record as a note (e.g., “city” in Figure 2 (b)), and the selected keywords are showcased on the right-hand side of the AR display. The selected keywords serve as quick notes, allowing users to express their thoughts and serve as reminders in a short period without recording complete sentences [1].

With the selected context keywords, the system can achieve within-context notes, as in [2, 3]. Users can also incorporate customized keywords on the left-hand side of the AR display, which are pre-defined by the users in advance, appearing when a keyword is selected. The default customized keywords are WH words, including what, why and how, and a question mark “?”. Users can select customized keywords to indicate whether a note is a question, and, if so, which type of WH or yes/no question it is (e.g., “What” in Figure 2 (c)).

To indeed accomplish beyond-context notes, keywords other than context keywords are required. In our brains, we usually think of something or some ideas when we are inspired by the things we see or hear, and we combine those things with our background knowledge, experiences and even personality to generate the thoughts. Therefore, when taking a beyond-context note and cannot find proper context keywords, users can opt to double-click on the most relevant context keyword (e.g., “rallies” in Figure 2 (d)) to generate derivative keywords via the LLM keyword derivation. The original keyword and 4 derivative keywords, a total of 5 items, are presented in a circular layout at the center of the AR display. Users can select derivative keywords that best match their intentions (e.g., “sign” in Figure 2 (e)), view more derivative keywords via the previous/next arrow buttons, or double-click on a derivative



Once a keyword is selected, the system automatically composes 3 candidate sentences via the LLM sentence organization using the selected keyword(s) as input, positioned on the bottom section of the AR display. Whenever a new keyword is selected, new candidate sentences using all selected keywords are automatically generated (Figure 2 (b) (c) (e)). Users can then select a candidate sentence that best matches their objectives as a note (e.g., “*What signs were displayed...*” in Figure 2 (f) (g)). However, if they cannot find any proper candidate sentence, they can double-click on any candidate sentence or selected keyword to skip the candidate sentence selection and directly record all selected keyword(s) (e.g., “*What, sign, city*” in Figure 2 (h) (upper)). Furthermore, when they prefer not to spend excessive time and attention on note-taking and need to take a note hastily without reading candidate sentences or even waiting for sentence generation, they have the option to select only context (or also customized) keywords and record these as a *quick*

In general, users primarily select context keywords and may also select a customized keyword(s), and then select a candidate

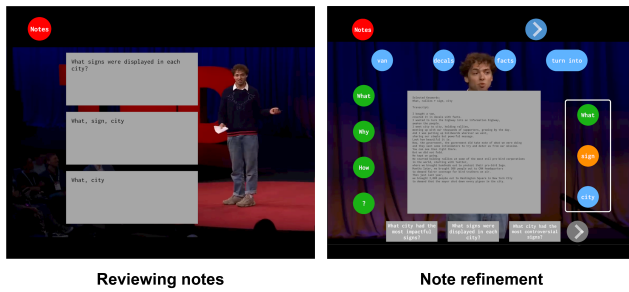


Figure 3: Reviewing notes by pressing the red “Notes” button and further reviewing transcripts and/or refining the notes.

sentence as a note or directly record selected keywords as a note. Furthermore, they can select only context (or also customized) keywords to record these as a quick note in just two or three selection steps, which means that a *quick note* is defined as one recorded only from (a) (b) or (a) (b) (c) steps in Figure 2 in GazeNoter. Although taking notes while staying engaged in the ongoing activity could cause distraction, GazeNoter allows users to freely employ their strategies to take notes based on the density of information in the speech-based activity. Instead of complete sentences, quick notes with only context (or also customized) keywords could be recorded in high-density information parts of the speech-based activity to avoid missing content. Furthermore, two types of historical information are preserved, allowing users to review them in less stressed parts. First, all previous context keywords can be viewed using the previous/next arrow buttons. Second, in note refinement, all selected keywords and their transcripts are also recorded. Therefore, users can add notes even if they have missed some content. Notably, since notes automatically generated by current LLMs could be comprehensive without any miss and could be integrated in our system in the future, our goal is to precisely generate notes match users’ intentions for both within-context and beyond-context notes.

3 Application

We demonstrate GazeNoter in two different scenarios: the static scenario, where the user is stationary and listening to a speech, and the mobile scenario, where the user is walking and participating in a meeting.

In the static scenario, to simulate attending speeches, TED Talks videos were used. Users sat on a chair and listened to speeches displayed on the AR headset, with the videos fixed in a designated space in the real world to mimic real speech scenarios. We encouraged them to explore every feature of GazeNoter, and asked them to think about their quick note-taking strategies for GazeNoter. They watched a TED Talks video and performed note-taking for questions in the Q&A sessions or important points and ideas they wanted to record. After the speeches, they could review the notes, see the transcripts of the notes, or even perform refinement in a 30-second time window, as a small time window between a speech and the Q&A session. Certainly, these could be performed during the speeches.

We further intended to showcase its performance in a mobile scenario, specifically in the context of walking meetings. Instead

of sitting on a chair, an introducer and a user walked around a room measuring $3.5\text{m} \times 4\text{m}$ for walking meetings. The introducer initiated a topic and spent 3 to 5 minutes describing it, followed by a discussion with the user during walking meetings. Each walking meeting was about 10 minutes. The user could freely take notes throughout the description and discussion. However, the user might need to review the notes for discussion with the introducer. This increased the mental pressure and required more urgent real-time note taking. In addition, walking meetings involve interactive communication and occasional face-to-face discussions, resulting in frequent eye contact and less time to take notes. Hence, the mental pressure in this scenario could be higher compared to listening to speeches in the previous scenario.

4 Conclusion

We propose a real-time note-taking system, GazeNoter, by integrating a user-in-the-loop LLM system with gaze selection on an AR headset to generate notes that are both within-context and beyond-context, matching the users’ intentions. We demonstrate GazeNoter in two different scenarios: the static one where the user is stationary and listening to a speech, and the mobile one where the user is walking and participating in a meeting, to showcase its performance, and how it reduces distraction and cognitive strain.

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